

Exercise for Chapter 4 and 5

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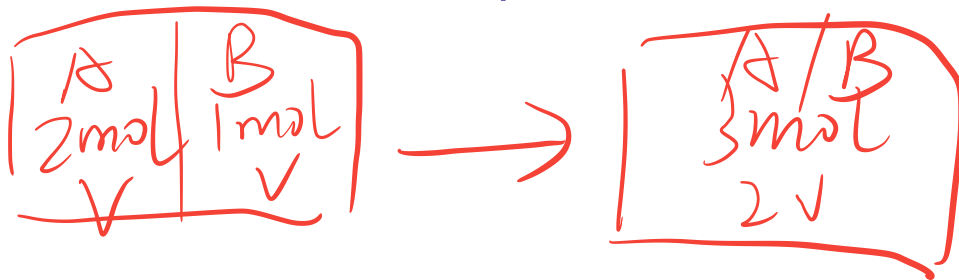
Chapter 4

4.1. A rigid container is divided into two compartments of equal volume by a partition.

One compartment contains 1 mole of ideal gas *A* at 1 atm, and the other contains 1 mole of ideal gas *B* at 1 atm. Calculate the increase in entropy which occurs when the partition between the two compartments is removed.

If the first compartment had contained 2 moles of ideal gas *A*, what would have been the increase in entropy when the partition was removed?

Calculate the corresponding increases in entropy in each of the preceding two situations if both compartments had contained ideal gas *A*.



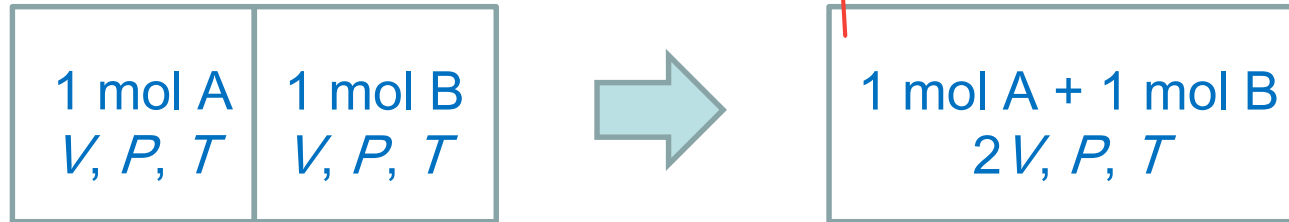
$$\begin{aligned}\Delta S &= S_A + S_B \\ &= 2 \ln \frac{2V}{V} + 1 \ln \frac{2V}{V} \\ &= \ln \frac{16}{9} + \ln \frac{2}{3}\end{aligned}$$

$\Rightarrow 0$

Chapter 4

Solution to 4.1

(1)

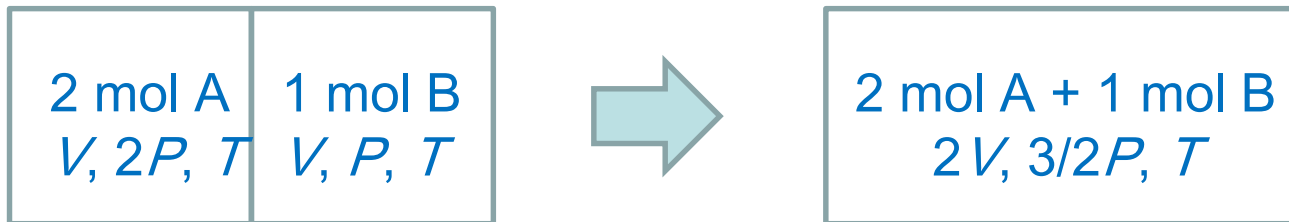


The volume occupied by 1 mol of A doubles $\therefore \Delta S_A = R \ln 2$

The volume occupied by 1 mol of B doubles $\therefore \Delta S_B = R \ln 2$

$$\therefore \Delta S_{\text{Total}} = \Delta S_A + \Delta S_B = 2R \ln 2$$

(2)



The volume occupied by 2 mol of A doubles $\therefore \Delta S_A = 2R \ln 2$

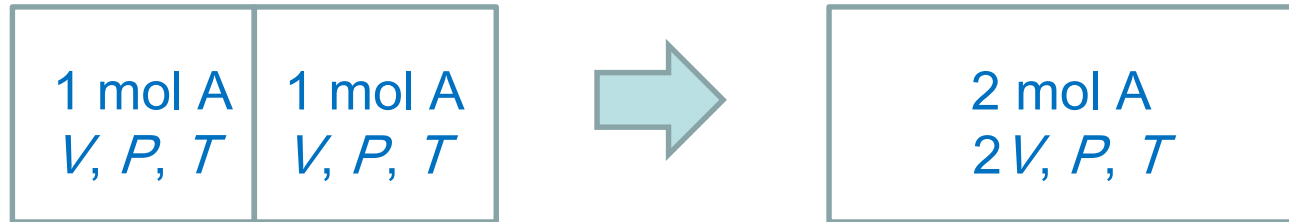
The volume occupied by 1 mol of B doubles $\therefore \Delta S_B = R \ln 2$

$$\therefore \Delta S_{\text{Total}} = \Delta S_A + \Delta S_B = 3R \ln 2$$

Chapter 4

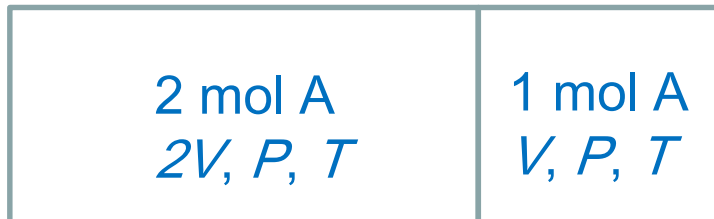
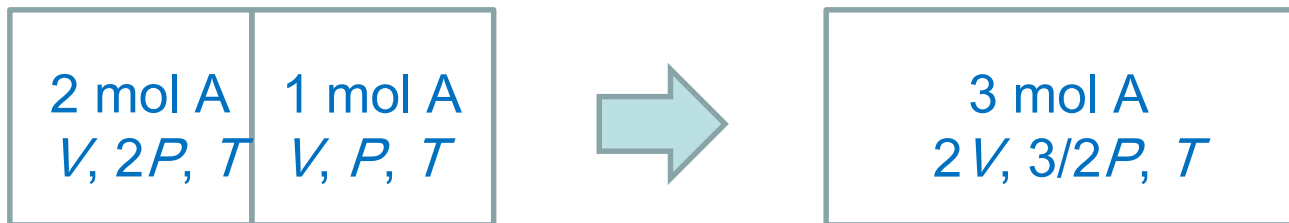
Solution to 4.1

(3)



$\Delta S = 0$ because the state of the gas doesn't change.

(4)



Double the volume of the 2 mol of A $\Delta S = 2R \ln 2$

Remove the partition $\Delta S = 0$

Decrease the volume of the 3 mol of A by the factor of $2/3$ $\Delta S = 3R \ln(2/3)$

$$\therefore \Delta S_{\text{Total}} = 2R \ln 2 + 3R \ln(2/3) = R \ln(32/27)$$

Chapter 4

4.2 Show that, when n atoms of A and n atoms of B form a randomly mixed solution, the fraction of the total number of distinguishable complexions which occur in the most probable distribution decreases with the increasing value of n .

Chapter 4

Solution to 4.2

The total number of distinguishable complexions is

$$\Omega_{\text{total}} = \frac{(2n)!}{n!n!} \quad (1)$$

the number of complexions which is the most probable distribution is

$$\Omega_{\text{max}} = \left[\frac{n!}{(0.5n)!(0.5n)!} \right] \left[\frac{n!}{(0.5n)!(0.5n)!} \right] \quad (2)$$

So the ratio of (2) to (1) is the fraction of the total number of distinguishable complexions which occur in the most probable distribution

$$f(n) = \frac{\Omega_{\text{max}}}{\Omega_{\text{total}}} = \frac{\left[\frac{n!}{(0.5n)!(0.5n)!} \right] \left[\frac{n!}{(0.5n)!(0.5n)!} \right]}{\frac{(2n)!}{n!n!}} = \frac{(n!)^4}{((0.5n)!)^4 (2n)!}$$

which decreases with the increasing value of n .

Chapter 4

4.4 On the assumption that copper–nickel alloys are random mixtures of copper and nickel atoms, calculate the mass of copper which, when mixed with 100 g of nickel, causes an increase in entropy of 15 J/K. The gram atomic weights of Cu and Ni are, respectively, 63.55 and 58.69.

Chapter 4

Solution to 4.4

Assume $n_{\text{Cu}} = \frac{m_{\text{Cu}}}{63.55\text{g/mol}} = n \text{ mol}$

$$n_{\text{Ni}} = \frac{100\text{g}}{58.69\text{g/mol}} = 1.70 \text{ mol}$$

$$X_{\text{Cu}} = \frac{n_{\text{Cu}}}{n_{\text{Cu}} + n_{\text{Ni}}} = \frac{n}{n + 1.70}$$

$$X_{\text{Ni}} = \frac{n_{\text{Ni}}}{n_{\text{Cu}} + n_{\text{Ni}}} = \frac{1.70}{n + 1.70} = 79.17\%$$

$$\Delta S_{\text{conf}} = -k_{\text{B}}(n_{\text{Cu}} + n_{\text{Ni}})(X_{\text{Cu}} \ln X_{\text{Cu}} + X_{\text{Ni}} \ln X_{\text{Ni}})$$

$$= -(n_{\text{Cu}} + 1.70) \times 8.3144 \times \left(\frac{n}{n + 1.70} \ln \frac{n}{n + 1.70} + \frac{1.70}{n + 1.70} \ln \frac{1.70}{n + 1.70} \right)$$

$$= 15 \text{ J/K}$$

Solve it, $n_{\text{Cu}}=1.025 \text{ mol}$, $m_{\text{Cu}}=65.14\text{g}$.

Chapter 4

4.5 We saw in Sec. 4.6 (Eq. 4.20) that $A = -nk_B T \ln Z$. We will see in chapter 5 that: $(\partial A / \partial T)_V = -S$. Perform the differentiation of A with respect to temperature and obtain the relationship for the entropy of an isolated system in terms of its partition function Z ; further obtain the relationship for the internal energy of an isolated system in terms of its partition function Z

Chapter 4

Solution to 4.5 $A = -nk_{\text{B}}T \ln Z$

$$\left(\frac{\partial A}{\partial T} \right)_V = -S = -nk_{\text{B}} \ln Z - nk_{\text{B}}T \frac{\partial \ln Z}{\partial T}$$

Thus: $S = nk_{\text{B}} \ln Z + nk_{\text{B}}T \frac{\partial \ln Z}{\partial T}$

Note that if we multiply both sides by T we obtain:

$$TS = nk_{\text{B}}T \ln Z + nk_{\text{B}}T^2 \frac{\partial \ln Z}{\partial T}$$

$$-nk_{\text{B}}T \ln Z = nk_{\text{B}}T^2 \frac{\partial \ln Z}{\partial T} - TS$$

Thus: $U = nk_{\text{B}}T^2 \frac{\partial \ln Z}{\partial T}$

Chapter 4

4.6 A weak magnetic field is applied to a system of up and down spins. The up spins have a slightly lower energy state than down spins. This is because the up spins are favored by the weak magnetic field. Thus,

$$\varepsilon^{\uparrow} < \varepsilon^{\downarrow}$$

a. Determine the partition function for this system under the influence of the weak magnetic field.

b. Determine the ratio of $\frac{n^{\uparrow}}{n^{\downarrow}}$ for very high temperatures and very low temperatures.

Chapter 4

Solution to 4.6

a.
$$Z = \sum \exp\left(\frac{-\varepsilon_i}{k_B T}\right) = \exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right) + \exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)$$

$$\frac{n^\uparrow}{n} = \frac{\exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right)}{\exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right) + \exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)} \quad \frac{n^\downarrow}{n} = \frac{\exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)}{\exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right) + \exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)}$$

Thus:
$$\frac{n^\uparrow}{n^\downarrow} = \frac{\exp\left(\frac{-\varepsilon^\uparrow}{k_B T}\right)}{\exp\left(\frac{-\varepsilon^\downarrow}{k_B T}\right)} = \exp\left(\frac{\varepsilon^\downarrow - \varepsilon^\uparrow}{k_B T}\right)$$

Chapter 4

Solution to 4.6

b.

$$\frac{n^{\uparrow}}{n^{\downarrow}} = \frac{\exp\left(\frac{-\varepsilon^{\uparrow}}{k_B T}\right)}{\exp\left(\frac{-\varepsilon^{\downarrow}}{k_B T}\right)} = \exp\left(\frac{\varepsilon^{\downarrow} - \varepsilon^{\uparrow}}{k_B T}\right)$$

$T \rightarrow \infty$, $n \rightarrow 1$, the numbers of up and down of are equal

The numerator of the exponential is positive: Thus when T decreases, the exponential becomes large.

As $T \rightarrow 0$, $n \rightarrow n^{\uparrow}$ and $n^{\downarrow} \rightarrow 0$

Chapter 5

5.1 Show that

$$\left(\frac{\partial S}{\partial V}\right)_P = \frac{c_p}{TV\alpha}$$

Solution:

$$dU = TdS - PdV \Rightarrow dS = \frac{dU}{T} + \frac{PdV}{T} \Rightarrow \left(\frac{\partial S}{\partial V}\right)_P = \frac{1}{T}\left(\frac{\partial U}{\partial V}\right)_P + \frac{P}{T}$$

$$U = H - PV \Rightarrow \left(\frac{\partial U}{\partial V}\right)_P = \left(\frac{\partial H}{\partial V}\right)_P - P$$

$$\left(\frac{\partial H}{\partial V}\right)_P = \left(\frac{\partial H}{\partial T}\right)_P \left(\frac{\partial T}{\partial V}\right)_P = \frac{c_p}{V\alpha}$$

$$\left(\frac{\partial S}{\partial V}\right)_P = \frac{1}{T}\left(\frac{\partial H}{\partial V}\right)_P$$

$$\left(\frac{\partial S}{\partial V}\right)_P = \frac{c_p}{V\alpha T}$$

Chapter 5

5.2 Show that

$$\left(\frac{\partial S}{\partial P}\right)_V = \frac{c_p \beta_T}{T \alpha} - V \alpha$$

Solution:

$$dU = TdS - PdV \Rightarrow \left(\frac{\partial S}{\partial P}\right)_V = \frac{1}{T} \left(\frac{\partial U}{\partial P}\right)_V$$
$$\left(\frac{\partial U}{\partial P}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V \left(\frac{\partial T}{\partial P}\right)_V \quad \left(\frac{\partial U}{\partial T}\right)_V = c_V \quad \left(\frac{\partial T}{\partial P}\right)_V = \frac{-\left(\frac{\partial V}{\partial P}\right)_T}{\left(\frac{\partial V}{\partial T}\right)_P} = \frac{\beta_T}{\alpha}$$

$$c_P - c_V = \frac{VT\alpha^2}{\beta_T} + \left(\frac{\partial S}{\partial P}\right)_V = \frac{c_V \beta_T}{T \alpha} \Rightarrow \left(\frac{\partial S}{\partial P}\right)_V = \frac{c_p \beta_T}{T \alpha} - V \alpha$$

Chapter 5

5.3 Show that

$$\left(\frac{\partial A}{\partial P}\right)_V = -\frac{S\beta_T}{\alpha}$$

Solution:

$$\left(\frac{\partial A}{\partial P}\right)_V = \left(\frac{\partial A}{\partial T}\right)_V \left(\frac{\partial T}{\partial P}\right)_V$$

$$\left(\frac{\partial T}{\partial P}\right)_V = \frac{\beta_T}{\alpha}$$

$$dA = \left(\frac{\partial A}{\partial T}\right)_V dT + \left(\frac{\partial A}{\partial V}\right)_T dV$$

$$dA = -SdT - PdV$$

$$\left(\frac{\partial A}{\partial T}\right)_V = -S$$

$$\left(\frac{\partial T}{\partial P}\right)_V = \frac{\beta_T}{\alpha}$$



$$\left(\frac{\partial A}{\partial P}\right)_V = -\frac{S\beta_T}{\alpha}$$

Chapter 5

5.4 Show that

$$\left(\frac{\partial A}{\partial V}\right)_P = -\left(\frac{S}{V\alpha} + P\right)$$

Solution:

$$\left(\frac{\partial A}{\partial V}\right)_P = \left(\frac{\partial A}{\partial T}\right)_P \left(\frac{\partial T}{\partial V}\right)_P = \left(\frac{\partial A}{\partial T}\right)_P \frac{1}{\alpha V}$$

$$dA = -SdT - PdV \quad \Rightarrow \quad \left(\frac{\partial A}{\partial T}\right)_P = -S - P\left(\frac{\partial V}{\partial T}\right)_P$$

$$\left(\frac{\partial A}{\partial V}\right)_P = (-S - P\alpha V) \frac{1}{\alpha V} = -\left(\frac{S}{\alpha V} + P\right)$$

Chapter 5

5.5 Show that

$$\left(\frac{\partial H}{\partial S}\right)_V = T \left(1 + \frac{V\alpha}{c_v\beta_T}\right)$$

Solution:

$$\left(\frac{\partial H}{\partial S}\right)_V = \left(\frac{\partial H}{\partial T}\right)_V \left(\frac{\partial T}{\partial S}\right)_V \quad \left(\frac{\partial T}{\partial S}\right)_V = \frac{T}{c_v}$$

$$H = U + PV \quad \Rightarrow \quad \left(\frac{\partial H}{\partial T}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V + V \left(\frac{\partial P}{\partial T}\right)_V = c_v + \frac{V\alpha}{\beta_T}$$



$$\left(\frac{\partial H}{\partial S}\right)_V = \frac{T}{c_v} \left[c_v + \frac{V\alpha}{\beta_T} \right] = T \left[1 + \frac{V\alpha}{c_v\beta_T} \right]$$

Chapter 5

5.6 Show that

$$\left(\frac{\partial H}{\partial V}\right)_S = -\frac{c_P}{c_v \beta_T}$$

Solution:

$$dH = TdS + VdP$$

Maxwell relations



$$\left(\frac{\partial H}{\partial V}\right)_S = V \left(\frac{\partial P}{\partial V}\right)_S$$

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$



From problem 5.1:

$$\left(\frac{\partial S}{\partial V}\right)_P = \frac{c_P}{TV\alpha}$$

$$\left(\frac{\partial P}{\partial V}\right)_S = -\frac{\left(\frac{\partial S}{\partial V}\right)_P}{\left(\frac{\partial S}{\partial P}\right)_V} = -\frac{c_P}{Vc_v \beta_T}$$

From problem 5.2:

$$\left(\frac{\partial S}{\partial P}\right)_V = \frac{c_v \beta_T}{T\alpha}$$

Chapter 5

5.7 Show that

$$\left(\frac{\partial c_P}{\partial P}\right)_T = -TV \left(\alpha^2 + \frac{d\alpha}{dT}\right)$$

Solution: $c_P = \left(\frac{\partial H}{\partial T}\right)_P \rightarrow \left(\frac{\partial c_P}{\partial P}\right)_T = \left(\frac{\partial}{\partial P} \left(\frac{\partial H}{\partial T}\right)_P\right)_T = \left(\frac{\partial}{\partial T} \left(\frac{\partial H}{\partial P}\right)_T\right)_P$

$$dH = TdS + VdP \rightarrow \left(\frac{\partial H}{\partial P}\right)_T = T \left(\frac{\partial S}{\partial P}\right)_T + V \quad \& \quad \left(\frac{\partial S}{\partial P}\right)_T = - \left(\frac{\partial V}{\partial T}\right)_P$$

$$\begin{aligned} \left(\frac{\partial c_P}{\partial P}\right)_T &= \left(\frac{\partial}{\partial T} \left[-T \left(\frac{\partial V}{\partial T}\right)_P + V \right]\right)_P = \left(\frac{\partial}{\partial T} [-T\alpha V + V]\right)_P \\ &= -\alpha V - TV \frac{d\alpha}{dT} - T\alpha^2 V + \alpha V = -TV \left(\alpha^2 + \frac{d\alpha}{dT}\right) \end{aligned}$$

Chapter 5

5.8 Show that

$$\left(\frac{\partial T}{\partial P}\right)_S = \frac{T\alpha V}{c_P}$$

Solution:

$$\left(\frac{\partial T}{\partial P}\right)_S = -\frac{\left(\frac{\partial S}{\partial P}\right)_T}{\left(\frac{\partial S}{\partial T}\right)_P} = -\frac{-\alpha VT}{c_P}$$

$$\left(\frac{\partial S}{\partial T}\right)_P = \frac{c_P}{T}$$

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P = -\alpha V$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \frac{\alpha VT}{c_P}$$

Chapter 5

5.9 Show that

$$\left(\frac{\partial P}{\partial V}\right)_S = -\frac{c_P}{c_v V \beta_T}$$

Solution:

$$\left(\frac{\partial P}{\partial V}\right)_S = -\frac{\left(\frac{\partial S}{\partial V}\right)_P}{\left(\frac{\partial S}{\partial P}\right)_V}$$

$$\left(\frac{\partial S}{\partial P}\right)_V = \left(\frac{\partial S}{\partial T}\right)_V \left(\frac{\partial T}{\partial P}\right)_V = -\frac{c_v}{T} \frac{\left(\frac{\partial V}{\partial P}\right)_T}{\left(\frac{\partial V}{\partial T}\right)_P} = \frac{c_v}{T} \frac{\beta_T V}{\alpha V} = \frac{c_v \beta_T}{T \alpha}$$

$$\left(\frac{\partial S}{\partial V}\right)_P = \left(\frac{\partial S}{\partial T}\right)_P \left(\frac{\partial T}{\partial V}\right)_P = \frac{c_P}{T} \frac{1}{\alpha V}$$



$$\left(\frac{\partial P}{\partial V}\right)_S = -\frac{c_P}{c_v V \beta_T}$$

Chapter 5

5.10 Show that

$$\left(\frac{\partial^2 G}{\partial P^2}\right)_T = -\frac{1}{\left(\frac{\partial^2 A}{\partial V^2}\right)_T}$$

Solution:

$$dG = -SdT + VdP \Rightarrow \left(\frac{\partial G}{\partial P}\right)_T = V \Rightarrow \left(\frac{\partial^2 G}{\partial P^2}\right)_T = \left(\frac{\partial V}{\partial P}\right)_T$$

$$dA = -SdT - PdV \Rightarrow \left(\frac{\partial A}{\partial V}\right)_T = -P$$

$$\left(\frac{\partial^2 A}{\partial V^2}\right)_T = -\left(\frac{\partial P}{\partial V}\right)_T = -\frac{1}{\left(\frac{\partial^2 G}{\partial P^2}\right)_T}$$

Chapter 5

5.11 Joule and Thomson showed experimentally that when a steady stream of nonideal gas is passed through a thermally insulated tube, in which is inserted a throttle valve, the temperature of the gas changes and the state of the gas is changed from P_1, T_1 to P_2, T_2 . Show that this process is isenthalpic. The change in T is described in terms of the Joule–Thomson coefficient, μ_{J-T} , as follows:

$$\mu_{J-T} = \left(\frac{\partial T}{\partial P} \right)_H$$

Show that

$$\mu_{J-T} = -\frac{V}{C_P} (1 - \alpha T)$$

and show that the Joule–Thomson coefficient for an ideal gas is zero.

Chapter 5

Solution to 5.11

- a. The process is adiabatic. Therefore $q = 0$ and work done by the gas, $w = P_2 V_2 - P_1 V_1$. Thus $\Delta H = q - w + (P_2 V_2 - P_1 V_1) = 0$. **isenthalpic**

b.

$$\mu_{J-T} = \left(\frac{\partial T}{\partial P} \right)_H \quad \left(\frac{\partial T}{\partial P} \right)_H = - \frac{\left(\frac{\partial H}{\partial P} \right)_T}{\left(\frac{\partial H}{\partial T} \right)_P} = - \frac{1}{c_P} \left(\frac{\partial H}{\partial P} \right)_T$$

$dH = TdS + VdP$


$$\left(\frac{\partial H}{\partial P} \right)_T = T \left(\frac{\partial S}{\partial P} \right)_T + V = -T \left(\frac{\partial V}{\partial T} \right)_P + V = -T\alpha V + V$$

$$\mu_{J-T} = - \frac{1}{c_P} (-T\alpha V + V) = \frac{V}{c_P} (T\alpha - 1)$$

Chapter 5

c.

$$\mu_{J-T} = -\frac{1}{c_p}(-T\alpha V + V) = \frac{V}{c_p}(T\alpha - 1)$$

For an ideal gas

$$\alpha = \frac{1}{T}, \quad \text{So} \quad \alpha T = 1 \quad \text{and} \quad \mu_{J-T} = 0$$

Chapter 5

5.12 Determine the values of $\Delta U'$, $\Delta H'$, $\Delta S'$, $\Delta A'$, and $\Delta G'$ for the following processes (in (c), (d), and (e), show that an absolute value of the entropy is required):

- a. The four processes in Problem 4.1
- b. The expansion of 1 mole of an ideal gas at the pressure P and the temperature T into a vacuum to double its volume
- c. The adiabatic expansion of 1 mole of an ideal gas from P_1 , T_1 to P_2 , T_2
- d. The expansion of 1 mole of an ideal gas at constant pressure from V_1 , T_1 to V_2 , T_2
- e. The expansion of 1 mole of an ideal gas at constant volume from P_1 , T_1 to P_2 , T_2

Chapter 5

Solution to 5.12

a. The four processes in Problem 4.1

A rigid container is divided into two compartments of equal volume by a partition. One compartment contains 1 mole of ideal gas A at 1 atm, and the other contains 1 mole of ideal gas B at 1 atm.

(1) when the partition between the two compartments is removed.

(2) If the first compartment had contained 2 moles of ideal gas A, when the partition was removed.

(3, 4) In each of the preceding two situations if both compartments had contained ideal gas A.

Chapter 5

Solution to 5.12

a. 1. $\Delta U = \Delta H = 0, \Delta S = R \ln 4, \Delta A = \Delta G = -T \Delta S = -RT \ln 4$

2. $\Delta U = \Delta H = 0, \Delta S = R \ln 8, \Delta A = \Delta G = -RT \ln 8$

3. $\Delta U = \Delta H = \Delta S = \Delta A = \Delta G = 0$

4. $\Delta U = \Delta H = 0, \Delta S = R \ln \left(\frac{32}{27} \right), \Delta A = \Delta G = -RT \ln \left(\frac{32}{27} \right)$

Chapter 5

Solution to 5.12

b. The expansion of 1 mole of an ideal gas at the pressure P and the temperature T into a vacuum to double its volume

b. $\Delta U = \Delta H = 0, \Delta S = R \ln(V_2 / V_1) = R \ln 2, \Delta A = \Delta G = -RT \ln 2$

c. The adiabatic expansion of 1 mole of an ideal gas from P_1 , T_1 to P_2 , T_2

c. $\Delta U = c_v (T_2 - T_1), \Delta H = c_p (T_2 - T_1), \Delta S = 0, \Delta A = \Delta U - S (T_2 - T_1),$
 $\Delta G = \Delta H - S (T_2 - T_1)$

Chapter 5

Solution to 5.12

d. The expansion of 1 mole of an ideal gas at constant pressure from V_1, T_1 to V_2, T_2

d. $\Delta U = c_v (T_2 - T_1), \Delta H = c_p (T_2 - T_1), \Delta S = c_p \ln(T_2 / T_1),$
 $\Delta A = \Delta U - S_1 (T_2 - T_1) - T_2 \Delta S, \Delta G = \Delta H - S_1 (T_2 - T_1) - T_2 \Delta S$

e. The expansion of 1 mole of an ideal gas at constant volume from P_1, T_1 to P_2, T_2

e. $\Delta U = c_v (T_2 - T_1), \Delta H = c_p (T_2 - T_1), \Delta S = c_v \ln(T_2 / T_1),$
 $\Delta A = \Delta U - S_1 (T_2 - T_1) - T_2 \Delta S, \Delta G = \Delta H - S_1 (T_2 - T_1) - T_2 \Delta S$

Chapter 5

5.13 Show that

$$1. TdS = c_V dT + \frac{T\alpha}{\beta_T} dV$$

$$2. TdS = c_P dT - TV\alpha dP$$

$$3. TdS = c_V \frac{\beta_T}{\alpha} dP + \frac{c_P}{\alpha V} dV$$

Chapter 5

Solution 5.13

a.

$$S = S(T, V)$$

$$\therefore dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV$$

under constant volume

$$TdS = \delta q_v = dU = c_v dT$$

$$\therefore \left(\frac{\partial S}{\partial T} \right)_V = \frac{c_v}{T}$$

according to the Maxwell relation

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V = - \left(\frac{\partial V}{\partial T} \right)_P \left(\frac{\partial P}{\partial V} \right)_T$$

$$\therefore TdS = c_v dT + \frac{T\alpha}{\beta_T} dV$$

Chapter 5

Solution 5.13

b.

$$S = S(T, P)$$

$$\therefore dS = \left(\frac{\partial S}{\partial T} \right)_P dT + \left(\frac{\partial S}{\partial P} \right)_T dP$$

under constant pressure

$$TdS = \delta q_p = dH = c_p dT$$

$$\therefore \left(\frac{\partial S}{\partial T} \right)_P = \frac{c_p}{T}$$

according to the Maxwell relation

$$\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P = -\alpha V$$

$$\therefore TdS = c_p dT - TV\alpha dP$$

Chapter 5

Solution 5.13

C.

$$S = S(P, V)$$

$$\therefore dS = \left(\frac{\partial S}{\partial V} \right)_P dV + \left(\frac{\partial S}{\partial P} \right)_V dP$$

$$\left(\frac{\partial S}{\partial V} \right)_P = \frac{\left(\frac{\partial S}{\partial T} \right)_P}{\left(\frac{\partial V}{\partial T} \right)_P} = \frac{\frac{c_P}{T}}{\alpha V} = \frac{c_P}{T \alpha V}$$

$$\left(\frac{\partial S}{\partial P} \right)_V = \left(\frac{\partial S}{\partial T} \right)_V \left(\frac{\partial T}{\partial P} \right)_V = \frac{c_V \beta_T}{T \alpha}$$

$$\therefore TdS = c_V \frac{\beta_T}{\alpha} dP + \frac{c_P}{\alpha V} dV$$

Chapter 5

5.14. Starting from the following equation valid for one mole of a closed system, obtain Maxwell relationships.

$$dH = TdS + VdP - \mu_0 VMd\aleph$$

Solution:

$$\left(\frac{\partial T}{\partial P}\right)_{S,\aleph} = \left(\frac{\partial V}{\partial S}\right)_{P,\aleph}$$

$$\left(\frac{\partial V}{\partial \aleph}\right)_{S,P} = -\mu_0 V \left(\frac{\partial M}{\partial P}\right)_{S,\aleph}$$

$$\left(\frac{\partial T}{\partial \aleph}\right)_{S,P} = -\mu_0 V \left(\frac{\partial M}{\partial S}\right)_{P,\aleph}$$

Chapter 5

5.15. Starting from the following equation valid for one mole of a closed system, obtain Maxwell relationships.

$$dA = -SdT - PdV + \mu_0 VHdM$$

Solution:

$$\left(\frac{\partial S}{\partial V} \right)_{T,M} = \left(\frac{\partial P}{\partial T} \right)_{V,M}$$

$$\left(\frac{\partial S}{\partial M} \right)_{T,V} = -\mu_0 V \left(\frac{\partial H}{\partial T} \right)_{M,V}$$

$$\left(\frac{\partial P}{\partial M} \right)_{V,T} = -\mu_0 V \left(\frac{\partial H}{\partial V} \right)_{M,T}$$

Chapter 5

5.16 The cycle shown in Figure 5.8 consists of two isotherms (AB and CD) and two isobars (BC and DA). Use the TdS equations to draw the T-S diagram for this cycle.

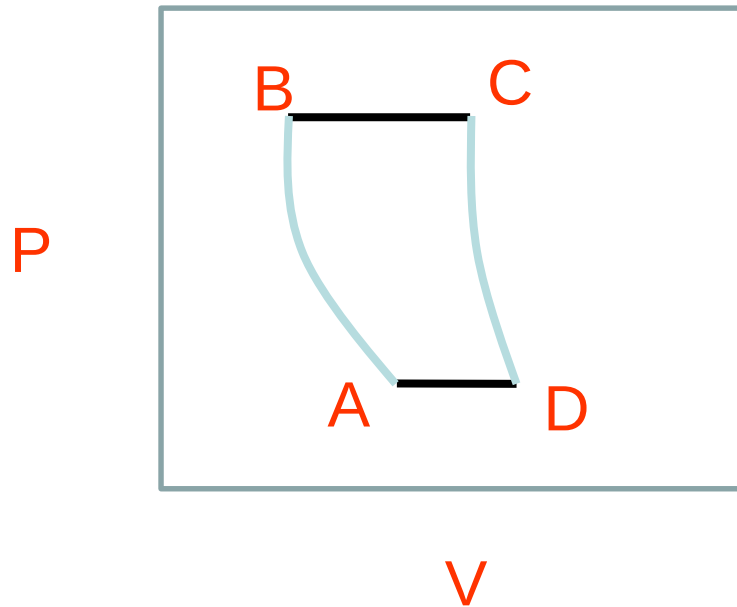


Figure 5.8 Pressure– volume diagram of two isothermals (AB and CD) and two isobaric changes (BC and DA) making up a cycle.

Chapter 5

Solution 5.16

Use the following TdS equations

$$1. TdS = c_V dT + \frac{T\alpha}{\beta_T} dV$$

$$2. TdS = c_P dT - TV\alpha dP$$

$$3. TdS = c_V \frac{\beta_T}{\alpha} dP + \frac{c_P}{\alpha V} dV$$

A→B from 1 at constant T, if V decreases, S decreases

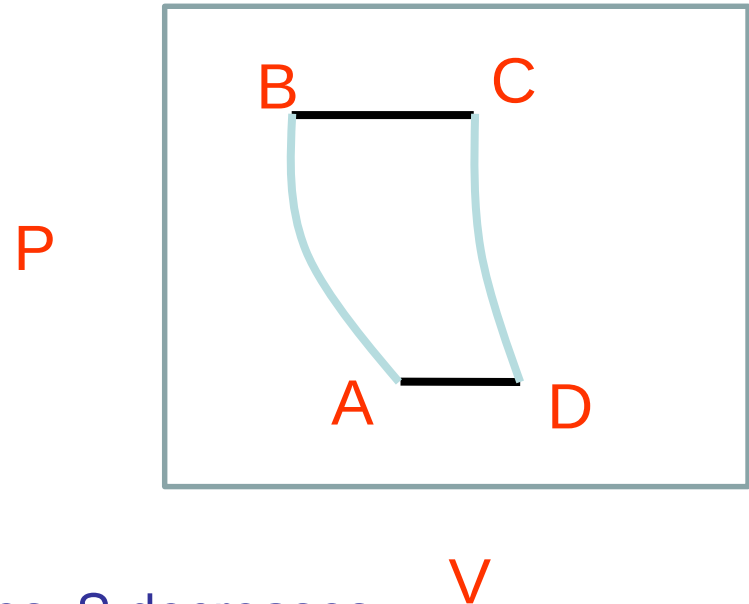
B→C from 3, at constant P, V increases then S increases.

From 2, at constant P if S increases T increases

C→D from 2, at constant T if P decreases then S increases

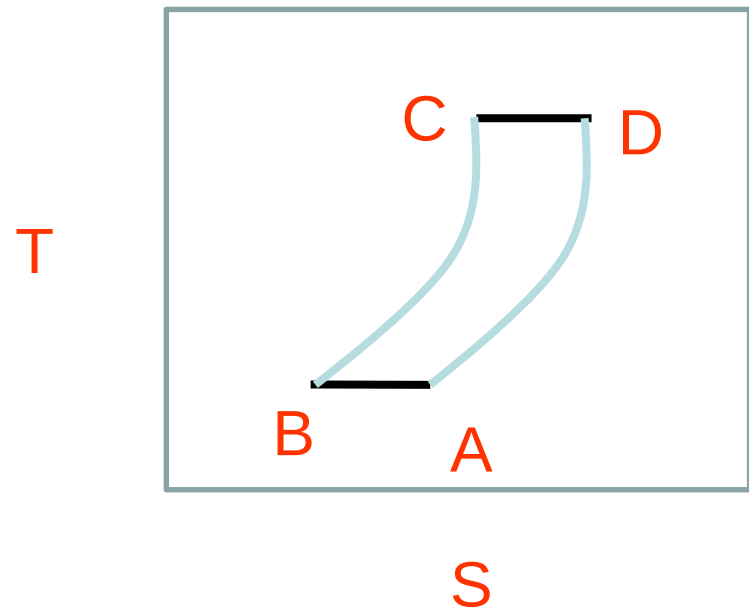
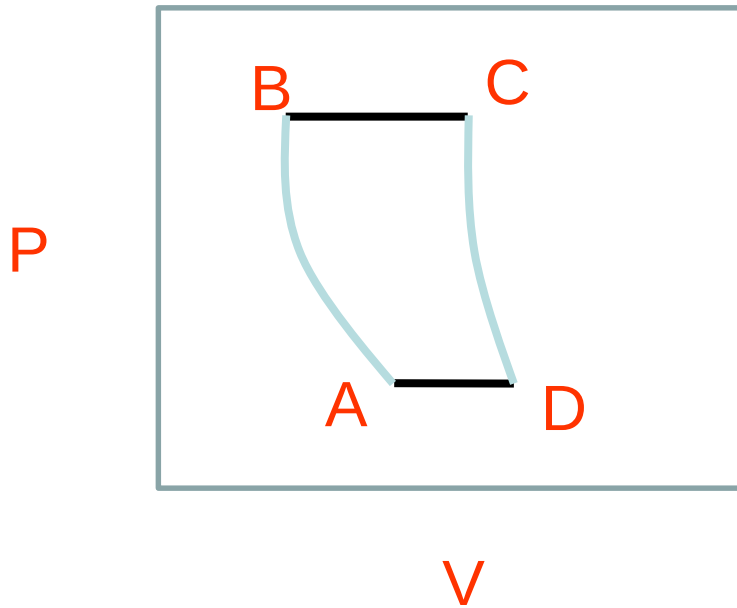
D→A from 3, at constant P, if V decreases S decreases

From 2, at constant P if S decreases, T decreases.



Chapter 5

Solution 5.16



Chapter 5

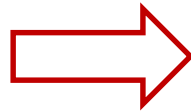
5.17. Show that: $\frac{\beta_T}{\beta_S} = \frac{c_P}{c_V} = \gamma$ $\beta_T \equiv -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$ $\beta_S \equiv -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_S$

Hint: use the TdS equations.

Solution: Start with the First and Second TdS equations:

$$\begin{aligned} 1. TdS &= C_V dT + \frac{T\alpha}{\beta_T} dV \\ 2. TdS &= C_P dT - TV\alpha dP \end{aligned}$$

Set $dS=0$



$$\begin{aligned} 1. C_V dT &= -\frac{T\alpha}{\beta_T} dV \\ 2. C_P dT &= TV\alpha dP \end{aligned}$$



$$\frac{C_P}{C_V} = -\frac{TV\alpha}{\frac{T\alpha}{\beta_T}} \left(\frac{\partial P}{\partial V} \right)_S = (V\beta_T) \frac{1}{V\beta_S} = \frac{\beta_T}{\beta_S}$$

Thermodynamics of Materials & Phase transformations

Review for Chapter 4 and 5

Chapter 4

1. A single macrostate of a system, which is determined when the independent variables of the system are fixed, contains a very large number of microstates, each of which is characterized by the manner in which

- The thermal energy of the system is distributed among the particles
- The particles are distributed in the configurations available to them

2. Although the occurrence of a system in any one of its microstates is equally probable (for the case of the closed system), greatly differing numbers of microstates occur in differing distributions. The distribution which contains the largest number of microstates is the most probable distribution.

Chapter 4

3. In real systems (with a very large number of particles) the number of microstates in the most probable distribution is significantly larger than the sum of all of the other microstates occurring in all of the other distributions. This most probable distribution is the equilibrium thermodynamic state of the system.

4. The relationship between the number of microstates available to the system, Ω , and the entropy of the system is given by the Boltzmann equation:

$$S = k_B \ln \Omega$$

in which k_B is Boltzmann's constant

Chapter 4

5. If a situation arises which allows an increase in the number of microstates available to the system (at constant energy), then spontaneous redistribution of the energy among the particles (or particles over the available configurations) occurs until the newly available most probable distribution occurs. The Boltzmann equation shows that an increase in the number of microstates made available to the system causes an increase in the entropy of the system.

6. The total entropy of a system is the sum of each of the aspects of entropy:

- Thermal entropy, S'_{th}
- Configurational entropy, S'_{conf}
- Any other aspect of entropy, such as spin entropy, electric dipole entropy, and so on

Chapter 4

7. The thermal entropy arises from the number of ways in which the thermal energy available to the system can be shared among the constituent particles, Ω_{th} .
8. The configurational entropy arises from the number of ways in which the particles can be distributed over the configurations available to them, Ω_{conf} .
9. The spin entropy arises from the number of ways in which the spins can be distributed over the sites available to them, Ω_{spin} .
10. Since any of the thermal distributions can be combined with any of the configurational distributions, the total number of microstates available to the system is the product $\Omega_{\text{th}} \Omega_{\text{conf}}$ ($\Omega_{\text{th}} \Omega_{\text{conf}} \Omega_{\text{spin}}$ if spins are present), and hence, from the logarithmic form of the Boltzmann equation, the total entropy of the system is the sum of the individual entropies of the system.

Chapter 4

✓ Partition function

$$P = \sum_{i=0}^{i=r} e^{-\beta \varepsilon_i} = e^{-\beta \varepsilon_0} + e^{-\beta \varepsilon_1} + e^{-\beta \varepsilon_2} + \dots + e^{-\beta \varepsilon_r} \quad i = 0, 1, 2, \dots, r$$

$$n_i = \frac{n}{P} e^{-\beta \varepsilon_i} \quad i = 0, 1, 2, \dots, r \quad \beta = \frac{1}{k_B T}$$

- ✓ From the point of view of microstates, an irreversible process is one which takes the system from **a less probable state** to **the most probable state**, and from the point of view of macrostates, an irreversible process takes the system from a **nonequilibrium state** to the **equilibrium state**.

Chapter 5

1. The Helmholtz free energy, A , is defined by $A \equiv U - TS$. In a closed system held at constant T and V , the Helmholtz free energy can only

- Decrease, for a spontaneous process
- Remain constant if the system is in equilibrium

Equilibrium at constant T and V is attained when the Helmholtz free energy achieves its minimum value.

2. The Gibbs free energy, G , is defined by $G \equiv H - TS$. In a closed system at constant pressure and temperature, the Gibbs free energy is a minimum at equilibrium. During an isothermal, isobaric process during which no form of work other than P - V work is performed (i.e., $w=0$), G can only

- Decrease for a spontaneous process
- Remain constant if the system is in equilibrium

Equilibrium is attained at constant T and P when the Gibbs free energy reaches its minimum value.

Chapter 5

3. For a change of state at constant pressure, $\Delta H = q_p$.
4. The chemical potential, μ_i , of the i th component can be expressed as

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T, P, n_j, j \neq i} = \left(\frac{\partial U}{\partial n_i} \right)_{S, V, n_j, j \neq i} = \left(\frac{\partial H}{\partial n_i} \right)_{S, P, n_j, j \neq i} = \left(\frac{\partial A}{\partial n_i} \right)_{S, V, n_j, j \neq i}$$

5. Internal energy varies with S , V , and composition as

$$dU = TdS - PdV + \sum_{i=1}^{i=k} \mu_i dn_i$$

Enthalpy varies with S , P , and composition as

$$dH = TdS + VdP + \sum_{i=1}^{i=k} \mu_i dn_i$$

Chapter 5

5. Helmholtz free energy varies with T , V , and composition as

$$dA = -SdT - PdV + \sum_{i=1}^{i=k} \mu_i dn_i$$

and Gibbs free energy varies with T , P , and composition as

$$dG = -SdT + VdP + \sum_{i=1}^{i=k} \mu_i dn_i$$

6. For one mole of a closed system with magnetic terms, the following holds.

$$dG = -SdT + VdP - V\mu_0 M d\mathfrak{H}$$

Chapter 5

7. The Maxwell relations for one mole of a closed simple system are

$$\begin{aligned}\left(\frac{\partial T}{\partial V}\right)_S &= -\left(\frac{\partial P}{\partial S}\right)_V & \left(\frac{\partial S}{\partial V}\right)_T &= \left(\frac{\partial P}{\partial T}\right)_V \\ \left(\frac{\partial T}{\partial P}\right)_S &= \left(\frac{\partial V}{\partial S}\right)_P & \left(\frac{\partial S}{\partial P}\right)_T &= -\left(\frac{\partial V}{\partial T}\right)_P\end{aligned}$$

8. With x , y , and z as state functions, the upstairs-downstairs-inside-out formula is

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$

Chapter 5

9. The Gibbs-Helmholtz equations are

$$\frac{d(G/T)}{dT} = -\frac{H}{T^2}$$

$$\frac{d(A/T)}{dT} = -\frac{U}{T^2}$$

Chapter 5

$$H \equiv U + PV$$

$$G \equiv H - TS$$

$$A \equiv U - TS$$

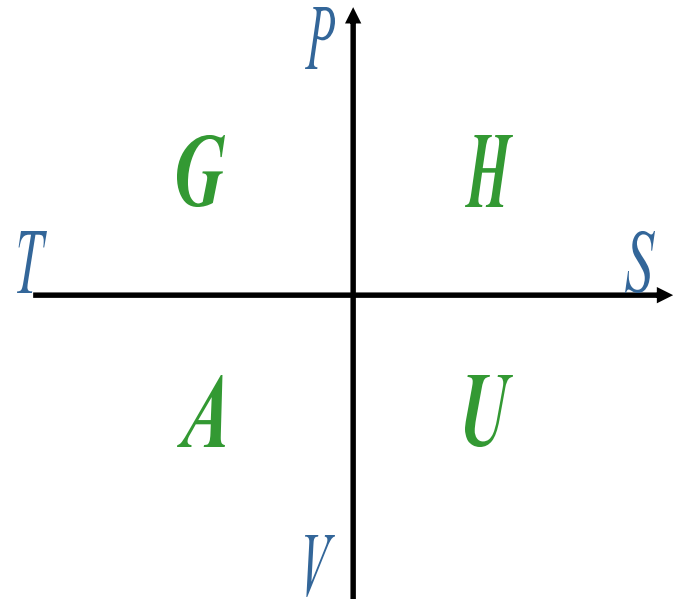
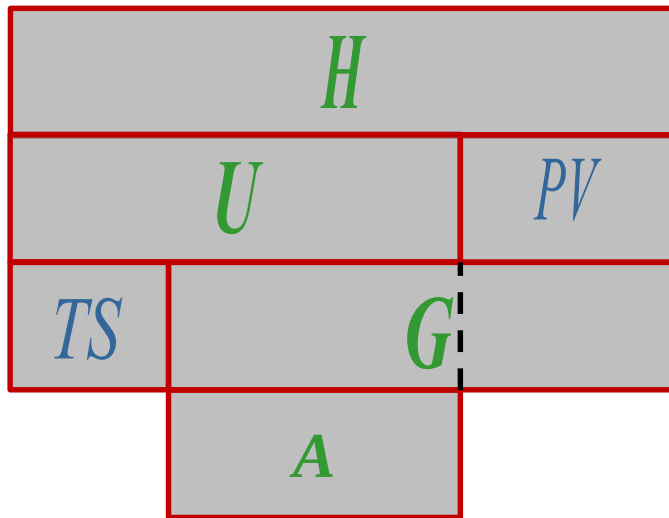
$$G \equiv A + PV$$

$$dU = TdS - PdV$$

$$dH = TdS + VdP$$

$$dA = -SdT - PdV$$

$$dG = -SdT + VdP$$



Thermodynamics of Materials & Phase transformations

The End

